MIL-B-36731
29 January 1971

MILITARY SPECIFICATION

BLOOD EXCHANGE-TRANSFUSION KIT, DISPOSABLE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a sterile, ready-to-use disposable, blood exchange-transfusion kit containing all equipment for use in blood exchange transfusion to combat Rh incompatibility effects.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical, General Specifications.
GG-N-196	Needles, Hypodermic, for Luer Syringes.
MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-plywood.
PPP-B-621	Box, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.
PPP-C-00186	Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.
PPP-T-76	Tape, Pressure-sensitive Adhesive Paper, Water-Resistant (for Carton Sealing).

FSC 6515

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Military

MIL-S-36175 Suit, Man S, Operating, Surgical (Shirt and Trousers).

STANDARDS

Federal

FED-STD-142 Parenteral Preparations

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

THE U. S. PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE: FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and General Regulations for its Enforcement.

(Applications for copies should be addressed to Food and Drug Administration, U.S. Department of Health, Education, and Welfare, Washington, D.C. 20203).

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, NATIONAL INSTITUTES OF HEALTH:

Public Health Service Regulation - Title 42 - Part 73.

(Applications for copies should be addressed to Food and Drug Administration, Washington, D.C. 20203).

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

A.N.S.I. Z70.1 American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, New York, N.Y. 10018).

3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber and plastic components. All rubber and plastic components shall be of the best quality and suitable in all respects for the intended purpose. All rubber and plastic components shall be sterilizable, free from pyrogens, compatible, and safe (nontoxic) when tested as specified in section 4. Those rubber and plastic components coming in contact with solutions being administered, and end protectors, shall contain not more than 5 parts per million (p.p.m.) of lead, barium and cadmium when tested as specified in 4.4.1. The plastics shall meet the requirements of the Pharmacopeia of the United States (U.S.P.) for Class II plastics and for Transfusion and Infusion Assemblies, when tested as specified in 4.4.2.

3.1.1.1 Rubber tubing. The rubber tubing shall be fabricated from natural rubber.

3.1.1.2 Plastic tubing. Plastic tubing shall be nontoxic, transparent, colorless polyvinyl. The tubing shall be flexible and pliable, but sufficiently rigid to prevent collapse during normal use. The tubing shall be free from all liquid lubricating substances which oxidize, exude, or sweat-out to the surface of the tubing.

3.1.1.3 Piercing device. The piercing device shall be fabricated from a suitable rigid plastic.

3.1.1.4 Needle adapter. The needle adapter shall be fabricated from a suitable rigid plastic.

3.1.1.5 End protectors. End protectors shall be fabricated from plastic tubes or sheaths.

3.1.2 Needles.

3.1.2.1 Cannulae of needles. Cannulae of needles shall be fabricated from 304 AISI stainless steel, and shall meet the corrosion resistance, elasticity, and resistance-to-breakage tests as specified in GG-N-196. The cannulae shall be fabricated from either seamless drawn, or welded and drawn tubing.

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3.1.2.2 Hubs. Hubs shall be class 3, type 302, 303, or 304, or class 8, alloy number 260, 270, or 360 of GG-I-526, or hubs shall be plastic class 8 (brass) hubs shall be nickel plated.

3.1.3 Metal slide clamp. The metal slide clamp shall be fabricated from aluminum. The clamp shall be compatible with the plastic tubing used to fabricate the upper tubing.

3.1.4 Other material. All other materials hereinafter specified shall be of the best quality and suitable in all respects for the purpose intended.

3.2 Style, design, and construction. Shall be a sealed kit containing a complete set of sterile items necessary for use in blood exchange transfusions. The kit shall be ready to use and disposable. The kit shall consist of a preformed, prepacked, plastic tray 15-1/2 inches long, 9-1/8 inches wide and 1-1/8 inches in height containing the following components:

Component	Quantity required
Airway needle, 18 gage	1
Blood transfusion set, with nylon blood filter, tubing, and control clamp.	1
Calcium gluconate ampul, 10%, 10cc.	1
Container for waste blood with drawstring closure, 2000 cc.	1
Exchange-transfusion record	1
Extension tube, 33 inch	1
Gauze sponge, 2 by 2 inch	3
Ruler, plastic, 15 cm.	1
Stopcock assembly (stopcock with 40" extension tube)	1
Syringe with 23 gage needle, 5cc.	1
Syringe with lock tip, 20cc.	2
Towel, antiseptic	1
Umbilical catheter, 5 Fr.	1
Umbilical catheter, 8 Fr.	1

3.2.1 Airway needle. Needle cannula shall be 18 gage with 1 inch length.

3.2.2 Blood transfusion set. The blood transfusion set shall consist of a piercing device, filter housing, nylon filter, 60 inches plastic tubing, 2-1/2 inches rubber tubing or proximal end (next to filter), 4 inches of rubber tubing on distal end (next to needle adapter), needle adapter and a flexible aluminum control clamp.

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3.2.2.1 Piercing device. The piercing device shall be suitable for the puncture of, and insertion into the port of a standard plastic blood bag, without damaging the bag, or into the blood bottle stopper shown in Figure 1, without exertion of undue pressure. The piercing device or the tip of the piercing device shall not break or fracture when placed perpendicular to the 1 cm. thick portion of a standard rubber stopper when gentle but firm pressure is applied. The construction of the piercing device shall be such as to prevent the retention of clots or fibrin. The piercing device shall make a snug, leakproof fit with the blood bottle stopper and the outlet ports of the plastic blood bag for the duration of the use of the blood transfusion set, and shall permit the blood transfusion set to be removed from the stopper or ports of the bag without difficulty and without damage to the stopper or the bag (that is, except for the puncture). The piercing device shall be ruggedly designed and structurally sound. It shall not break, or leak air or fluid during use and shall not be brittle or soft, nor become brittle or soft under normal storage conditions.

3.2.2.1.1 Durability of piercing device. The piercing device shall be tested for durability as specified in 4.4.3. The piercing device shall be so firmly attached to the filter housing that the puncture of the device and the housing shall not loosen or permit leaking, nor shall the device or housing become damaged. The piercing device shall be capable of being inserted into the blood containers (into the blood bottle rubber stoppers, Figure 1, and the outlet ports of the blood bag), as described in 4.4.3, without the device breaking or bending to the degree of causing difficulty when properly inserted into the blood container.

3.2.2.2 Filter housing. The filter housing shall be fabricated from transparent, substantially colorless, semi-rigid, suitable plastic. The housing shall be flexible enough to permit squeezing by hand, yet rigid enough to withstand the collapse test as specified in 4.4.4. The housing shall not leak or separate when tested as specified in 4.4.5. The filter housing shall be constructed to prevent the passage of clots or fibrin. Housing shall be constructed so that the fluid shall pass through the filter and through the output connector into the attached tubing. The filter shall be permanently secured to the housing and shall not permit fluid to pass between the internal wall of the housing and shall be closed at its bottom end. The filter material shall be nylon mesh, 70 to 100 mesh sieve (sieve number as defined in the U.S.P. for Standard Sieves) and shall have a filtering area of not less than 4 square inches.

3.2.2.3 Rubber tubing. Rubber tubing shall be attached to bottom of filter housing and to plastic tubing and shall be approximately 2-1/2 inches in length. At the end of the plastic tubing, a 4 inch length of rubber tubing shall be securely attached.

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EXCHANGE TRANSFUSION RECORD

NAME OF BABY

NAME OF MOTHER

BIRTH WEIGHT

BLOOD GROUP

TIME COMMENCING EXCHANGE

TIME FINISHING EXCHANGE

HOSPITAL NO.

DATE OF DELIVERY:

TIME

APGAR SCORE AT DELIVERY

INITIAL HEMOGLOBIN:

BILIRUBIN:

POST-EXCHANGE BILIRUBIN:

AGE OF BABY IN HOURS:

TIME	OUT		IN		PULSE	RESPIRATION	VENOUS PRESSURE	MEDICATION	COMMENTS
	Amount	Total	Amount	Total					

Figure 1. - Exchange Transfusion Record

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3.2.2.4 Needle adapter. The needle adapter shall be a rigid suitable plastic and so constructed as to be attached to the rubber tubing in a leakproof manner. The other end of the adapter shall be tapered to fit a needle having a standard Luer hub in accordance with ANSI Standard Z70.1 and shall form an airtight leakproof connection. The needle adapter shall be tested for dimensions as specified in 4.4.6. Needle adapters with Luer tapered ends which do not fit the test gage within specified limits shall be considered not acceptable. In addition, when the test gage is fitted onto the Luer tapered end, as specified in 4.4.6, there shall be no movement on the tapered end when the gage is pressed lightly from side to side in all directions.

3.2.2.5 Control clamp. The control clamp shall be constructed of flexible aluminum. Edges shall be smooth enough to prevent the clamp from cutting the tubing when in either the open or closed position. The action of opening and closing the clamp shall not cut the tubing when tested in accordance with 4.4.7.

3.2.3 Calcium gluconate. Shall be calcium gluconate injection, U.S.P. and shall be in accordance with all applicable requirements of FED-STD-142 and as stated herein.

S2 Classification. Shall be type I, class 1. Shall be suitable for intramuscular and intravenous use.

S5.2.1 The following additional requirements and tests are added to this paragraph:

Shall contain 10 percent (w/v) of Calcium Gluconate in each cc.of injection and a suitable stabilizer within the applicable assay limits of the injection.

S6.12.1 Aqueous vehicles. Shall be water for injection.

S6.3 Volume in final containers. The final container (ampul) shall contain 10.0 cc. plus the required excess.

S6.8.3 Nonpotency dated parenterals. In lines 5 and 6, delete "the date of delivery to the Government" and substitute "to its inclusion into the kit".

3.2.3.1 Ampules. Shall be in accordance with all applicable requirements of Interim Federal Specification PPP-C-00186A except as indicated herein:

Ampules. Shall comply with the following classification:

GROUP A CLASS 1 TYPE a STYLE 7 GRADE 1 or 2

CLOSURE K

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Para 3.2.3.1 Continued.

Labeling. Each ampule shall be labeled in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act, and shall include the information required below, however, the information need not appear in sequence indicated.

(a) the item name designated as "CALCIUM GLUCONATE INJECTION U.S.P."

(b) the quantity of contents designated as 10 ml.

(c) the percentage of calcium gluconate as "10%"

(d) the lot or control number

(e) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(f) the name and percentage of stabilizer.

(g) the statement "See package insert for dosage" or similar statement

(h) the route of administration.

3.2.3.2 Brochure. A circular brochure or other printed matter shall be packaged within package, setting forth as a minimum: A description of the item, its clinical use and contraindications.

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3.2.4 Container for waste blood. The container shall be made of polyethylene and have a volume capacity of 2000 ± 50 cc. The container shall be imprinted with a volumetric scale graduated in 100 cc. increments (100 through 2000). Graduations shall be accurate within ± 50 cc. This scale shall be legibly imprinted with a nontoxic ink compounded to bond to the surface and assure against loss of legibility due to handling or ordinary contact with standard solutions.

3.2.5 Exchange transfusion record. Each kit shall be provided with an exchange transfusion record in accordance with Figure 1.

3.2.6 Extension tube. The extension tube shall be terminated at one end by attachment to a tapered male adapter and a female adapter at the other end. The adapter shall be attached at each end to the plastic tubing in a leak-proof manner.

3.2.6.1 Taper of male Luer adapter. The adapter shall be provided with a standard Luer taper in accordance with ANSI Standard Z70.1-1955, and shall make a leakproof connection with a standard Luer hub cavity constructed in accordance with Figure 3 of ANSI Standard Z70.1-1955 for a hypodermic needle. This tapered end shall be tested for dimensions as specified in 4.4.6.

3.2.6.2 Diameter and depth of cavity of female Luer adapter. The female adapter shall make a secure watertight union with male Luer slip connectors. The female adapter shall be tested for depth and diameter as specified in 4.4.8.

3.2.7 Gauze sponge. The gauze sponge shall be 12 ply, approximately 2 inches by 2 inches when folded.

3.2.8 Ruler, plastic. The plastic ruler shall be 15 cm. long and shall be permanently marked in 1 cm. increments. Graduation shall be accurate within ± .75 cm.

3.2.9 Stopcock assembly. The stopcock shall have two side ports (female connectors), placed at 90° to each other, each having internal female Luer tapers and a rotating plug whose axis of rotation is perpendicular to the plane of the inlet-outlet ports. The third port of the stopcock shall have a female Luer connector. The rotating plug shall be of contrasting color to the stopcock body. The rotating plug shall be provided with a projecting arm which when superimposed over a body outlet port shall indicate that flow path through the stopcock. The upper end of the arm shall contain a male Luer lock. The opposite end of this Luer lock shall be fitted with a rubber cap of such consistency as to enable it to reseal after being punctured with a needle. Attached to the male Luer taper shall be a 40 inch length of plastic tubing. The extension tube shall have securely attached a female Luer taper fitting at one end and a male Luer taper fitting at the other end.

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3.2.9.1 Male and female Luer connectors. The male connector shall have a tapered tip to receive a standard hypodermic needle. The tip shall make a mechanically stable, fluid tight connection (see 3.2.10) with a hypodermic needle having a Luer hub with hub cavity in accordance with Figure 2 (female end) of A.N.S.I. Z70.1. The female connector shall have a tapered cavity to receive a standard hypodermic syringe. The cavity of the female connector shall make a mechanically stable, fluid tight connection (see 3.2.10) with a syringe having a Luer tip in accordance with Figure 2 (male end) of A.N.S.I. Z70.1.

3.2.9.2 Female Luer lock connector. The female Luer lock connector shall be in accordance with Figure 3 of A.N.S.I. Z70.1.

3.2.10 Leakage and separation of set. The stopcock assembly shall be tested for leakage and separation as specified in 4.4.9.

3.2.11 Syringe with 23 gage needle, 5cc. Shall be a 5cc. disposable plastic syringe with needle. Syringe shall contain a standard male Luer hub, and needle shall contain a standard female Luer hub. The needle shall be 23 gage with 1 inch length. The material, style, design, dimensions and construction shall be in accordance with MIL-S-36175 for Type II, size 3, except the packaging requirements of that specification shall not apply.

3.2.12 Syringe, with lock tip, 20cc. Shall be a 20cc. disposable, plastic syringe with standard Luer lock tip. The material, style, design, dimensions and construction shall be in accordance with MIL-S-36175 for Type II, size 3, except the tip shall be in accordance with Figure 4 of A.N.S.I. Z70.1 and the packaging requirements of that specification shall not apply.

3.2.13 Towel, antiseptic. The towel shall be antiseptic and supplied in a foil package.

3.2.14 Umbilical catheter, 5 Fr. The catheter shall be a 5 Fr. open end umbilical catheter with two eyes and an open tip. The proximal end shall have permanently attached a standard female Luer adapter. The female adapter shall be approximately 13/16 inch in overall length and shall have a tapered cavity to receive a standard hypodermic syringe. The cavity of the female adapter shall make a mechanically stable, fluid tight connection with a syringe having a Luer tip in accordance with Figre 2 (male end) of A.N.S.I. Z70.1. The female adapter shall be tested for depth and diameter in accordance with 4.4.8. The catheters shall be marked at 6, 7, and 8 centimeters from the tip.

3.2.15 Umbilical catherter, 8 Fr. Shall be as specified in 3.2.13 except the catheter shall be a 8 Fr. size.

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3.3 Sterility. All components of the kit shall be sterile. When tested as specified in 4.4.10, the airway needle, blood transfusion set, extension tube, stopcock assembly, syringes and catheters shall comply with the sterility test requirements of the U.S.P.

3.4 Toxicity. When tested as specified in 4.4.11, the airway needle, blood transfusion set, extension tube, stopcock assembly, syringes and catheters shall comply with the requirements for safety (toxicity) of the U.S.P.

3.5 Animal implantation. When tested as specified in 4.4.12, the catheters shall comply with the requirements of the Implantation Test, Plastic Containers, of the U.S.P.

3.6 Freedom from pyrogens. When tested as specified in 4.4.13, the fluid path, end connectors, needles, and catheters shall be pyrogen free. All the above shall be in accordance with the additional standards: Whole Blood (Human) of the Public Health Service Regulation - Part 73.

3.7 Certificates of sterility, freedom from pyrogens, safety, biological tests and animal implantation tests. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested and found to be sterile, free from pyrogens, nontoxic (safe), and have passed the biological tests and the animal implantation test. This certificate shall include the specific tests performed and the tests results. The record of the animal implantation test and its results shall be certified by the pathologist who performed or supervised performance of the animal implantation test. For destination inspection, certificates shall be submitted by the contractor directly to Defense Personnel Support Center, Directorate of Medical Materiel, Technical Operations Division, 2800 South 20th Street, Philadelphia, Pa. 10101, Attention: Quality Assurance Branch. One copy of each shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via the government inspector.

3.8 Manufacturer's records. The manufacturer shall place a lot control number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made available for inspection by any duly qualified medical officer of the Armed Forces, National Institutes of Health, any official of the Federal Food and Drug Administration, or any duly authorized Government representative. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

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3.9 Compatibility. The materials of which the sets are manufactured shall not react with, or be acted upon by the liquids with which they will be in contact during normal use and operation, nor shall there be any materials incorporated in the set which will cause any deleterious effect when the set is used for the purpose intended.

3.10 Instructions. Each set shall be furnished with a list of the contents of the pack and procedural instructions for use (except for actual insertion of the catheter).

3.11 Workmanship. Workmanship shall be first class throughout. All materials, treatments, and procedures used in the fabrication and assembly of the sets shall be such that the finished sets shall not show any evidence pf decomposition, deterioration, or other conditions which may adversely affect the appearance or serviceability of the sets at time of delivery or during storage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specifiedd herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastic, rubber, metal, and cotton, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination shall be one **blood transfusion set.**

Table I. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

Inspection lot shall be the same as sterilization lot unless otherwise specified.

4.2.2 For tests.

4.2.2.1 For sterility test. Sampling for the sterility test shall be in accordance with the United States Pharmacopeia (U.S.P.) for parenteral devices.

4.2.2.2 For pyrogen, and safety tests. Sampling procedures for pyrogen, and safety tests shall be conducted as indicated in the U.S.P., for Transfusion and Infusion Assemblies.

4.2.2.3 For lead test. Samples shall be taken from each lot of rubber or plastic as indicated in 4.4.

4.2.3 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II.

Table II. Sampling for Other Tests

Component (unit of product)	Characteristic	Req.	Test Procedure	Insp. level	AQL (percent defective)
Plastic components	Material - Class II	3.1	4.4.1	**	*
	Plastic, lead (Biological)	3.1	4.4.2	**	*

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Table II. Sampling for Other Tests (Cont'd)

Component (unit of product)	Characteristic	Req.	Test Procedure	Insp. level	AQL (percent defective)
Blood Transfusion Set	Piercing device durability	3.2.2.1	4.4.3	S-3	*
	Filter housing collapse	3.2.2.2	4.4.4	S-3	*
Filter housing	Leakage and separation	3.2.2.2	4.4.5	S-3	*
Needle adapter	Dimensions and taper	3.2.2.4	4.4.6	S-3	*
Control clamp	Metal slide clamp	3.2.2.5	4.4.7	S-3	*
Adapter	Diameter and depth of cavity	3.2.6.2 3.2.14	4.4.8	S-3	*
Blood Transfusion Set	Leakage and separation	3.2.10	4.4.9	S-3	*
Set	Sterility	3.3	4.4.10	***	*
Set	Safety	3.4	4.4.11	***	*
Catheter	Animal implantation	3.5	4.4.12	***	*
Set	Freedom from pyrogen	3.6	4.4.13	***	*

* Acceptance number shall be zero for all sample sizes.
** Sufficient sample units to perform the biological tests shall be selected from each lot.
*** Sampling for tests shall be on component lots.

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4.2.4 Sample unit usage. Where possible, the sample unit shall be used for determination of two or more test characteristics.

4.3 Examination. The sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table III. Inspection, however, shall not be restricted to the classified possible defects listed in Table III.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Set incomplete, one or more components missing.
102	Plastic, rubber, or metal components not smoothly finished; not free of fits, blisters, scratches, tears, cracks, nicks, bubbles, roughness, tackiness, bloom or crazing, as applicable.
103	Components not free of foreign matter.
104	Waste container and towels not free from defects such as cuts and tears.
105	Labeling of calcium gluconate not as specified.
106	Catheter not symmetrical in shape.
107	Inside of catheter not free of obstructing material.
108	Catheter eyes not smooth and free of flash.
Minor	
201	Packaging, packing and marking not as specified.
202	Instructions missing, incomplete, illegible.
203	Poor workmanship (not resulting in a major defect).

4.4 Tests.

4.4.1 Etractable metals. A 2 gram composite sample shall be obtained from the rubber and plastic components. The material shall be cut into small slivers, so that no thickness exceeds 3 mm., and eluted for 72 hours at 50° C., 24 hours at 70° C., or 60 minutes at 121° C. in two mentrua: 0.9% sodium chloride and cottonseed oil. Each gram of the material shall be covered with 5 milliliters (ml.) of the menstruum. The menstrua are collected, agitated after elution, and then submitted to spectographic analysis for lead, barium, and cadmium. The metals are reported in parts per million. Thus 1.0 ppm in elute is equivalent to extracting 1.0 mg. from 200 g test material dispersed in 1,000 ml. of elution menstruum.

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4.4.2 Biological tests. The plastics shall be tested in accordance with the U.S.P. for Class II plastics.

4.4.3 Piercing device durability test. The piercing device (on assembled set) shall be inserted into and removed from a blood bottle as shown on Figure 2 and inserted and twisted into and removed from a plastic blood bag (FSN 6515-926-4767), in the normal manner. This operation shall be repeated twice. While the piercing device or tip of the piercing device is being inserted into the stopper perpendicular to the 1 centimeter thick portion of the standard rubber stopper, a gentle but firm pressure shall be applied to the set. The piercing device shall then be immersed in Anticoagulant Solution "A" in accordance with current revision of the Pharmacopeia of the United States and the Additional Standards: Whole Blood (Human) of the Public Health Service Regulations - Part 73, for 4 hours. The piercing device shall then be removed from the solution, and immediately inserted in a plastic blood bag as specified above.

4.4.4 Filter and drip chamber collapse test. The filter chamber shall be filled to one-quarter its capacity, the drip chamber to one-third its capacity, and the recipient tube shall be completely filled with water at 100° F. Close off the entrance to the piercing connector lumen. Hold set in a vertical position, with needle adapter three feet below the bottom of the drip chamber. Remove end protector from needle adapter and allow fluid to flow from open end of set. Subject to this condition for one minute.

4.4.5 Leaking of set test. Samples of recipient sets (sterilized) shall be subjected to a compressive force between the flange of the piercing device and the filter chamber (i.e., normal to flange of piercing device) of 25 pounds for 2 seconds. A torque of 4 inch pounds shall be exerted for 2 seconds between the piercing device and the filter chamber. A separation force of 8 pounds for 2 seconds shall be exerted between the piercing device and recipient tube. The set shall then be subjected to an internal pressure (hydrostatic pressure) of 15 pounds per square inch gage (p.s.i.g.) for 30 seconds. This test shall be performed with a gage as shown in Figure 8 of A.N.S.I. Standard Z70.1-1955 (and having a suitable fluid path throughout the length of the gage) fitted to the Luer tapered tip of the needle adapter.

4.4.6 Needle adapter. The tapered end of the needle adapter shall be tested for proper dimensions and taper by means of the ring gage shown in Figure 7 of A.N.S.I. Standard Z70.1-1955. This gage shall be slipped onto the tip with moderate pressure.

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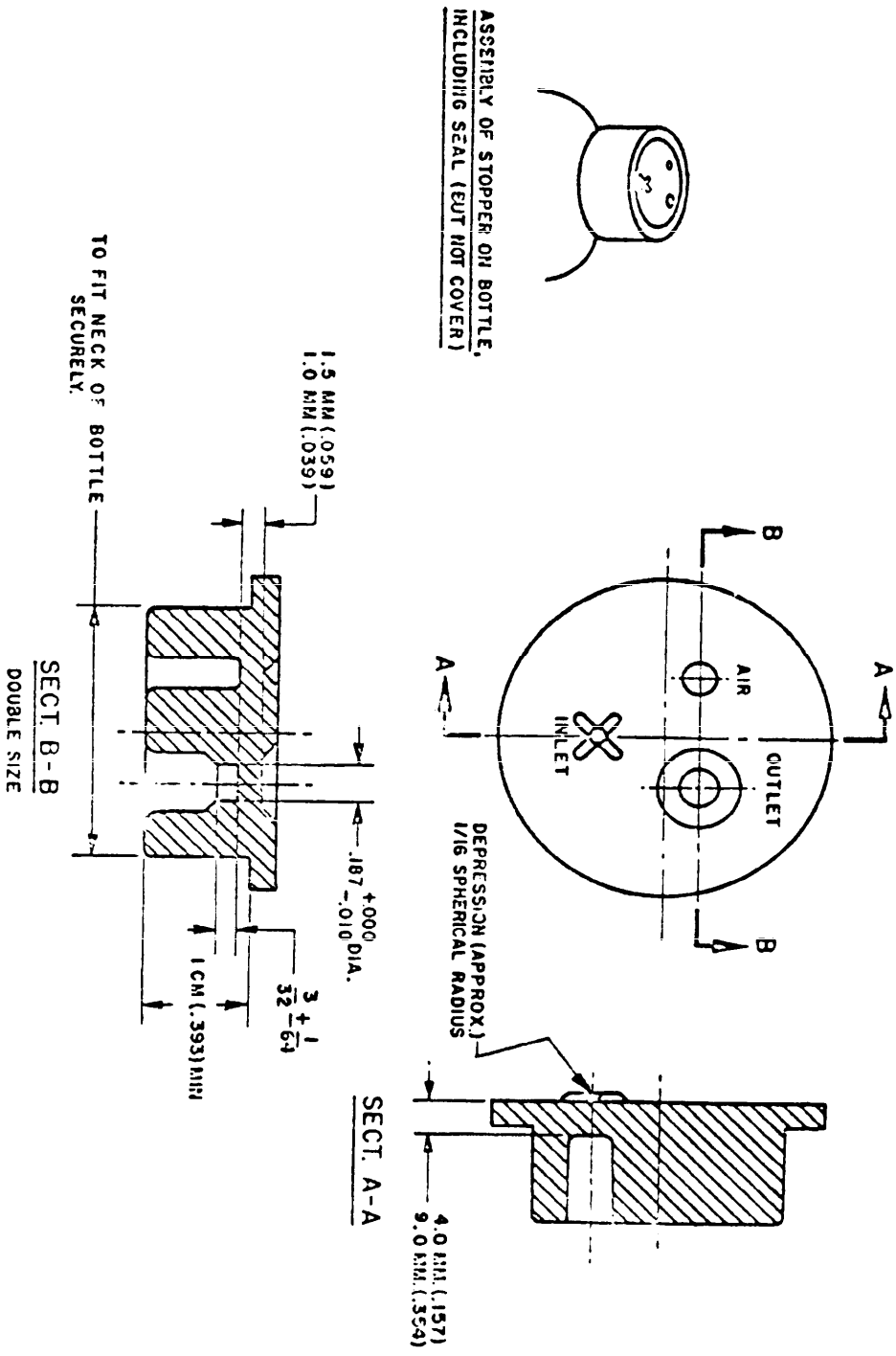


FIG. 2

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4.4.7 Metal clamp test. The metal clamp shall be operated for 10 complete cycles from stop flow to full flow with the clamp located approximately in center of tubing length while subjected to an internal pressure (hydorstatic pressure) of 4 p.s.i.g. At the completion of the test the tubing shall be examined for cuts.

4.4.8 Diameter and depth of cavity of female Luer adapter. The cavity of the female Luer adapter shall be tested on the male Luer taper gage in accordance with plug gage as shown on Figure 5 of A.N.S.I. Standard Z70.1-1955.

4.4.9 Leakage and separation of stopcock assembly. Samples of finished assembly after sterilization shall be subjected to an internal air pressure of 8 pounds per square inch (p.s.i.). This test shall be performed with a gage shown on Figure 8 of A.N.S.I. Standard Z70.1-1955 (and having a suitable fluid path throughout the length of the gage) fitted to the Luer tapered tip of the needle adapter and gage as shown on Figure 5 of A.N.S.I. Standard Z70.1-1955 (and having a suitable fluid path throughout the length of the gage) fitted to the female Luer connectors. This pressure shall be maintained for 30 seconds while set is completely submerged under water. The set shall be observed for leakage as indicated by air bubbles.

4.4.10 Sterility test The kits shall be tested for sterility in accordance with U.S.P. The testing for sterility shall be performed on at least the blood transfusion set, needles, extension tube, stopcock assembly, syringes, catheters, and towels, as representative of the sterility of all the set components.

4.4.11 Safety (toxicity) test. Samples of the finished kits shall be tested for safety, in accordance with the U.S.P., for transfusion and infusion assemblies.

4.4.12 Animal implantation test. Samples of catheters shall be tested in accordance with the U.S.P. for Implantation Test, Plastic Containers, except after "72 hours" add "and not to exceed 144 hours.", and delete "Examine macroscopically the area of the tissue surrounding the center portion of each implant strip. Use a magnifying lens if necessary.", and substitute: "The muscle tissue surrounding the implants shall be examined both grossly and microscopically by a pathologist certified by the American Board of Pathology or the American College of Veterinary Pathologists."

4.4.13 Freedom from pyrogens. Samples of the finished kits shall be tested for freedom from pyrogens, in accordance with the U.S.P., for transfusion and infusion assemblies.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One blood exchange transfusion kit, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	5 units	40 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Components comprising each blood exchange transfusion kit shall be contained in individual cells of a preformed plastic tray 15-1/2 inches long, 9-1/8 inches wide and 1-1/8 inches high. The tray with components shall be wrapped in a 28 x 28 inch tissue drape interlined with polypropylene and package in a box of appropriate size and design constructed in accordance with PPP-B-676, except that commercial colors are acceptable. The box shall be further wrapped with polypropylene film and heat sealed. The unit shall be packaged in such a manner as to maintain sterility of the tray and contents unless the package is damaged or opened.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. In addition to any commercial markings, each unit package shall be marked in accordance with MIL-STD-129 and the requirements of the Federal Food, Drug and Cosmetic Act. Lot (control) number, contract or purchase order number and name of contractor shall be shown. The method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry Heat Sterilization) shall be shown following the level of packing and date of packing. Example: "A-1/70-G." Final sterilization shall be on or after indicated packaging date. In addition, marking shall include the legend: "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED."

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number and name of contractor shall be shown. The method of sterilization as specified in 5.3.1 shall also be shown.

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5.3.3 Exterior containers. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number and the method of sterilization as specified in 5.3.1 shall also be shown.

NOTE: The item identification shall be shown on the unit, intermediate, and exterior container as follows:

"BLOOD EXCHANGE-TRANSFUSION KIT DISPOSABLE"

6. NOTES

6.1 Intended use. The blood exchange-transfusion kit is intended for use in blood exchange transfusions when speed and safety are essential to combat the effects of Rh incompatibility.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Federal stock number.

6.3 Federal Stock Number (FSN) coverage. This specification is intended to cover the following item appearing in the Federal Supply Catalog: FSN 6515-837-7491 Blood Exchange-Transfusion Kit, Disposable. It does not include all types, sizes, or grades as indicated by the title or all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1550

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-B-36731 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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